

g·SENSOR

8 FNIRS

INSTRUCTIONS FOR USE





Instructions for Use: g.SENSOR 8 fNIRS
Version 1.20.01

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1 IMPORTANT NOTES

1.1 THE INTENDED FUNCTION OF THE EQUIPMENT

The g.SENSOR 8 FNIRS device is intended to be used in research and scientific investigations for the combined measurement of hemodynamic responses associated with neural behaviour with low voltage electro-physiological signals like EEG. The device must not be used for medical applications nor for treatment or diagnosis in medical environments.

The device must not be used for the determination of brain death.



ATTENTION

- You must not look directly at the optical sensors when the device is switched on.
- Do not remove or reconfigure the sensor setup when the device is switched on.
- The g.SENSOR 8 FNIRS is not protected against liquids! When cleaning the device, always make sure that no liquids can enter the device!



WARNING

The device is powered via a lithium-ion accumulator. The accumulator must only be replaced by an accumulator provided via the manufacturer.

1.2 SPECIAL TREATMENT OF OPTODES AND CABLES

g.tec uses special lightweight, thin and highly flexible cables for the optical sensors to provide high comfort and easy cap mounting, especially for multi-channel recording of FNIRS and EEG. These cables are sensitive and need to be treated with special care. Following some basic guidelines will prolong the lifetime of optodes and cables:

- Never pull on optode cables.
- Avoid knots in cables.
- Do not put cables and optical sensor parts and the optode connector box into any liquid.
- Avoid exposure to direct sunlight or chemical agents.
- Do not autoclave optodes or the optode connector box.
- Make sure that no gel remains on optodes or cables after cleaning when used together with wet EEG electrodes.
- Protect from contamination with gel, water or disinfectant.
- Always make sure that optodes, cables, connector box and caps are completely dry before storing.
- Don't cut, kink or pinch optode cables; light bending is safe.

1.3 WARRANTY/REPLACEMENT OF OPTODES

Optode and cable lifetime depends on proper usage, careful treatment and cleaning, and appropriate storage. g.tec will provide warranty replacement only if there is no visible physical damage to the parts, such as: damaged, broken or pinched cables; or damaged housings or connectors.

1.4 LIMITATIONS OF REPROCESSING FOR OPTODES

- Do not use any other detergent as mentioned in this policy!
- Do not perform automated reprocessing in Washer Disinfectors (WD) or Endoscope Washer Disinfectors (EWD)!
- Do not machine-wash!
- Do not use a laundry dryer or other hot air devices!
- Do not put into an ultrasonic bath!
- Do not autoclave optodes!

For disinfection use disinfection wipes such as Mikrozid® sensitive wipes #70000807, Mikrozid® AF wipes #109203 from Schülke & Mayr GmbH, www.schuelke.com, or similar products.

1.5 HARDWARE REQUIREMENTS OF PC/NOTEBOOK

An additional USB 2.0 port for the Bluetooth Dongle is required.

2 g.SENSOR 8 FNIRS INTRODUCTION

g.SENSOR 8 FNIRS is g.tec's portable multimodal device with wireless technology that allows combined recordings of EEG and FNIRS signals (together with a g.tec biopotential amplifier). The EEG and FNIRS signals are recorded synchronously under the g.tec MATLAB-Simulink platform. The g.SENSOR 8 FNIRS system acquires 8 low-power transmitters for forehead FNIRS recordings or 8 high-power transmitters for central/motor area FNIRS recordings. The transmitters are LED based and emit light at nominal standard wavelengths of 760 and 850 nm.

The 8 transmitters and 2 receivers result in 2 x 4 FNIRS channels and are based on the FNIRS technology of Octamon+ from Artinis (www.artinis.com). For detailed information about the Octamon+ hardware, please see the Octamon+ manuals.

The optode connector box is positioned in the back of the head, allowing EEG recordings according to the extended 10/20 system in parallel to the acquisition of FNIRS.



3 AVAILABLE OUTPUT SIGNALS OF G.SENSOR 8 FNIRS

g.SENSOR 8 FNIRS provides the following output signals, in addition to the EEG output channels that are transmitted to the PC/laptop, and can be viewed and recorded, depending on the used software:

- Optical densities output signals for all 8 channels and both wavelengths (odd channels for 760 nm and even channels for 850 nm wavelengths)
- Concentrations: HbO (oxy-haemoglobin) and HbR (deoxy-haemoglobin) signals for all 8 channels (odd channels = HbO, even channels = HbR)
- Sample count: increasing numbers of samples acquired from the FNIRS module
- Signal quality for each channel (0 = bad signal quality, 1 = good signal quality)

g.SENSOR 8 fNIRS



4 COMPONENTS

Item	Amount
Optode connector box for 10 detachable optical sensor with cable connected battery housing	1
Bluetooth dongle	1
Transport case	1
Low power optodes (transmitter)	8
High power optodes (transmitter)	8
Optode receiver	2
Optode spacer for low power	2
Optode spacer for high power	2
Optode holder rings	20
Lithium-Ion battery	2
Charging device for battery	1

The USB-Bluetooth dongle, which comes with the system, guarantees a good wireless connection between the device and the computer.

USB Bluetooth dongle



5 MARKINGS ON THE g·SENSOR 8 FNIRS

Top view

Bottom view



Labels	Description
	CE mark
	Do not dispose with domestic waste. Dispose it via the separate collection system for electrical and electronic equipment.
	Applied part BF
	Safety class II
	Follow instructions for use
FNIRS SN 16122	Serial number
ID 000666F1EE28	Bluetooth identification number
IP20	IP 20: Protected against objects > 12.5 mm.

S1

S1: permanent operation

T1, ..., T8

Connection socket for Transmitter 1 to Transmitter 8

R1, R2

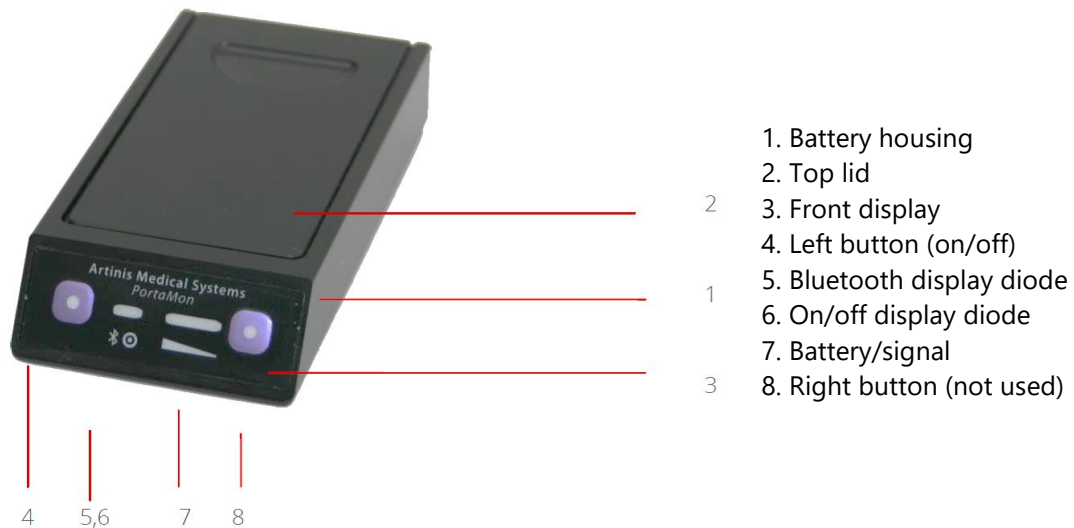
Connection socket for Receiver 1 and Receiver 2

The battery housing is connected to the optode connector box via a non-detachable cable. It includes the removable and rechargeable accumulator and the Bluetooth transceiver.

Do not put the box inside a metallic compartment during FNIRS recording, as this could affect the wireless data transmission.

Top view

Bottom view



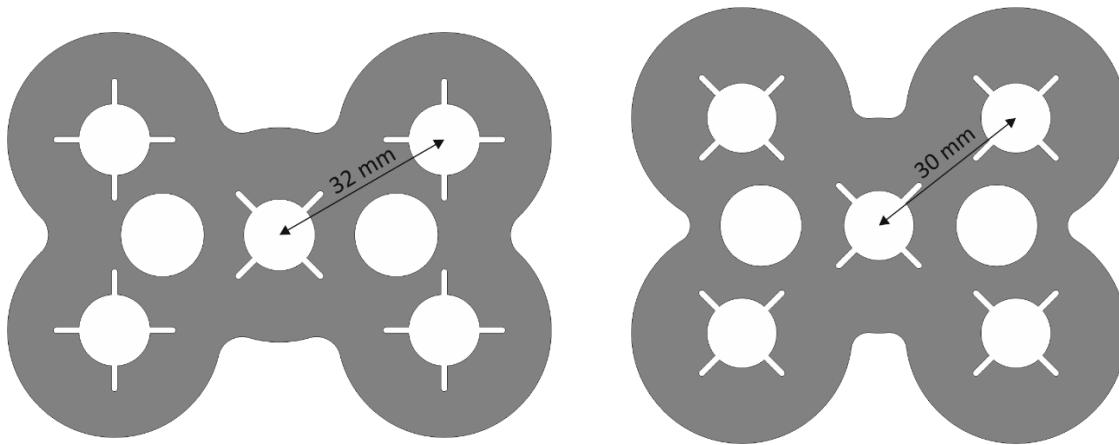
6 OPTODES AND EEG CAP

6.1 OPTODE SPACERS

The system includes 2 optode spacer pads for the low-power transmitters and 2 optode spacer pads for the high-power transmitters. The pads are mounted in the electrode cap with special optode holder rings and guarantee a precise positioning with correct distances between the transmitters and the receiver. The spacer pads are made of a black bio-compatible rubber, which also provides ambient light shielding around the optodes. The receiver optodes (R1, R2) are always used in the center positions of the spacer pads.

Optode spacer for low-power transmitters

Optode spacer for high-power transmitters



6.2 OPTODE HOLDER RINGS

The optode holder rings allow to adjust the position level of the optodes. Usually, the lowest level is used to assure a good contact of the tip of the optode to the skin. In case the optode causes pain to the subject, a higher level may be better. Press the button on the side of the holder ring to adjust the level or to release the optode from the cap.

Holder rings attached to the optode spacer



6.3 OPTODES

Transmitters (low-power and high-power) and receivers have very similar housings. High-power optodes are marked with "HP" and low-power optodes with "LP", respectively. The receiver is always placed in the middle position of the spacer pad.

Low-power transmitter



High-power transmitter



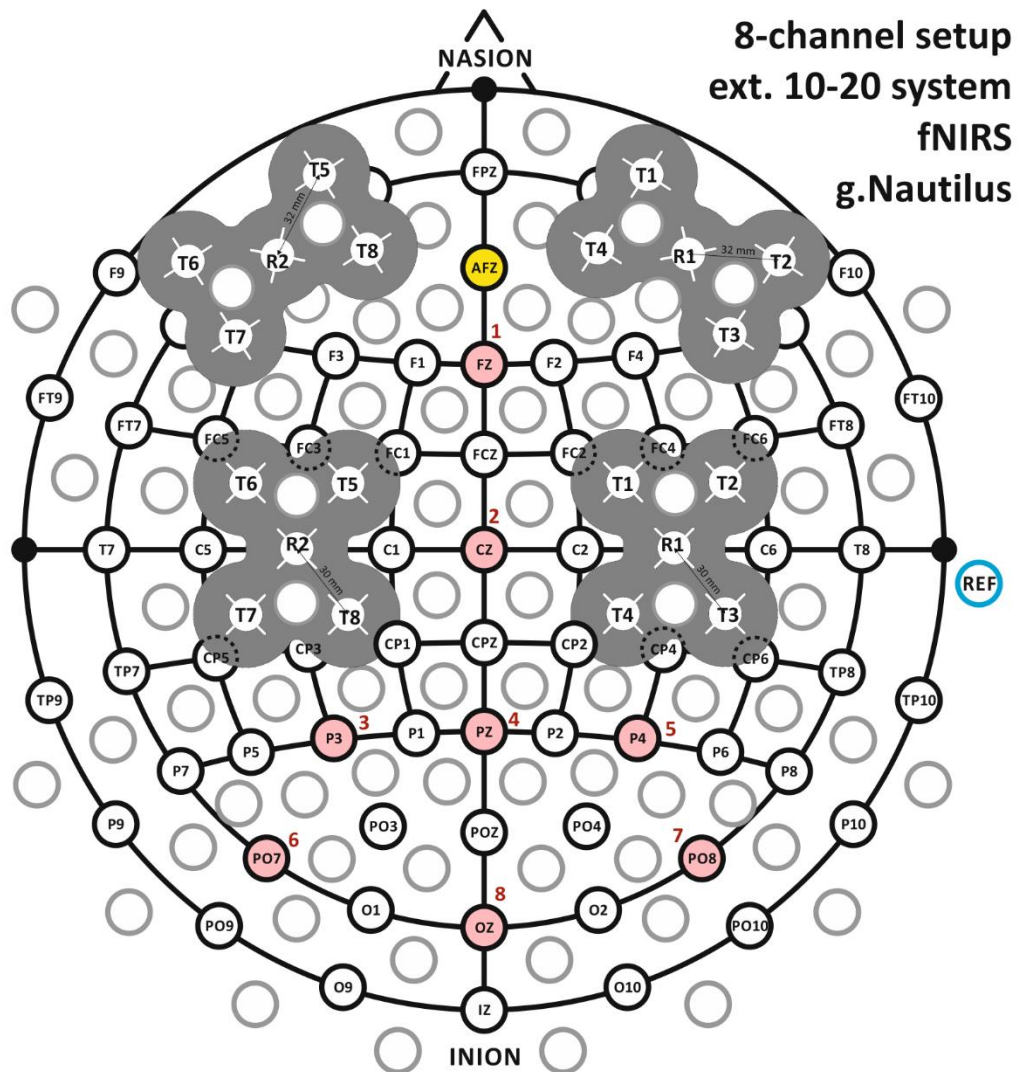
Receiver

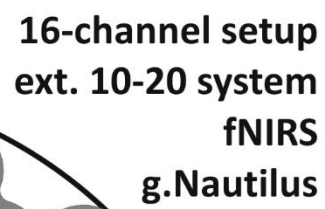


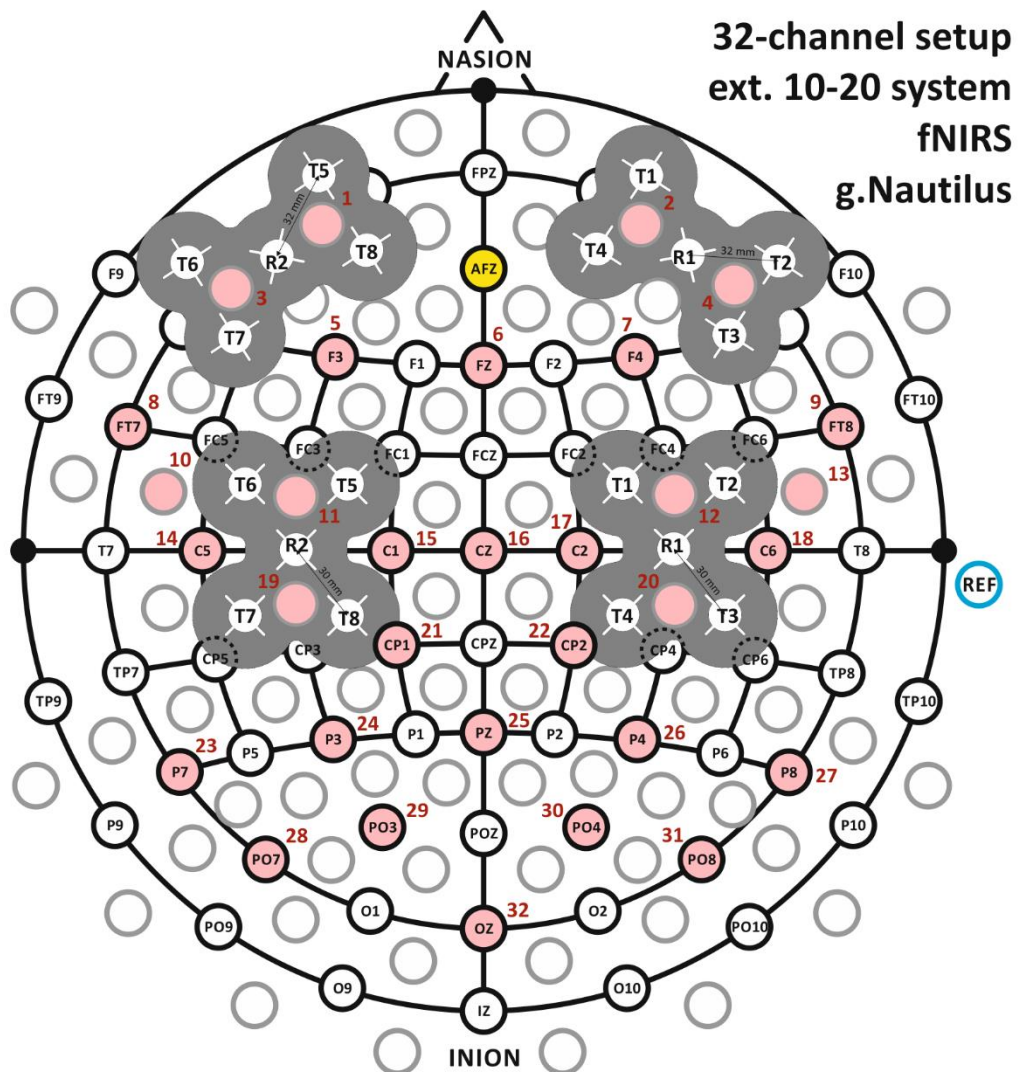
6.4 EEG ELECTRODE AND OPTODE POSITIONS

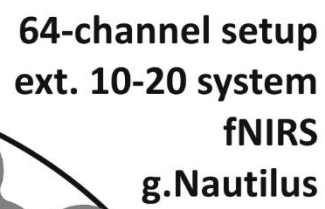
Due to the specific placement of the FNIRS optodes and spacer pads, the standard positions of the EEG electrodes are modified for the 16-, 32- and 64-channel setup. However, electrode positions and optode spacer incl. optode positions, can be changed by the user freely.

Assembling 8 channels, 8 fNIRS

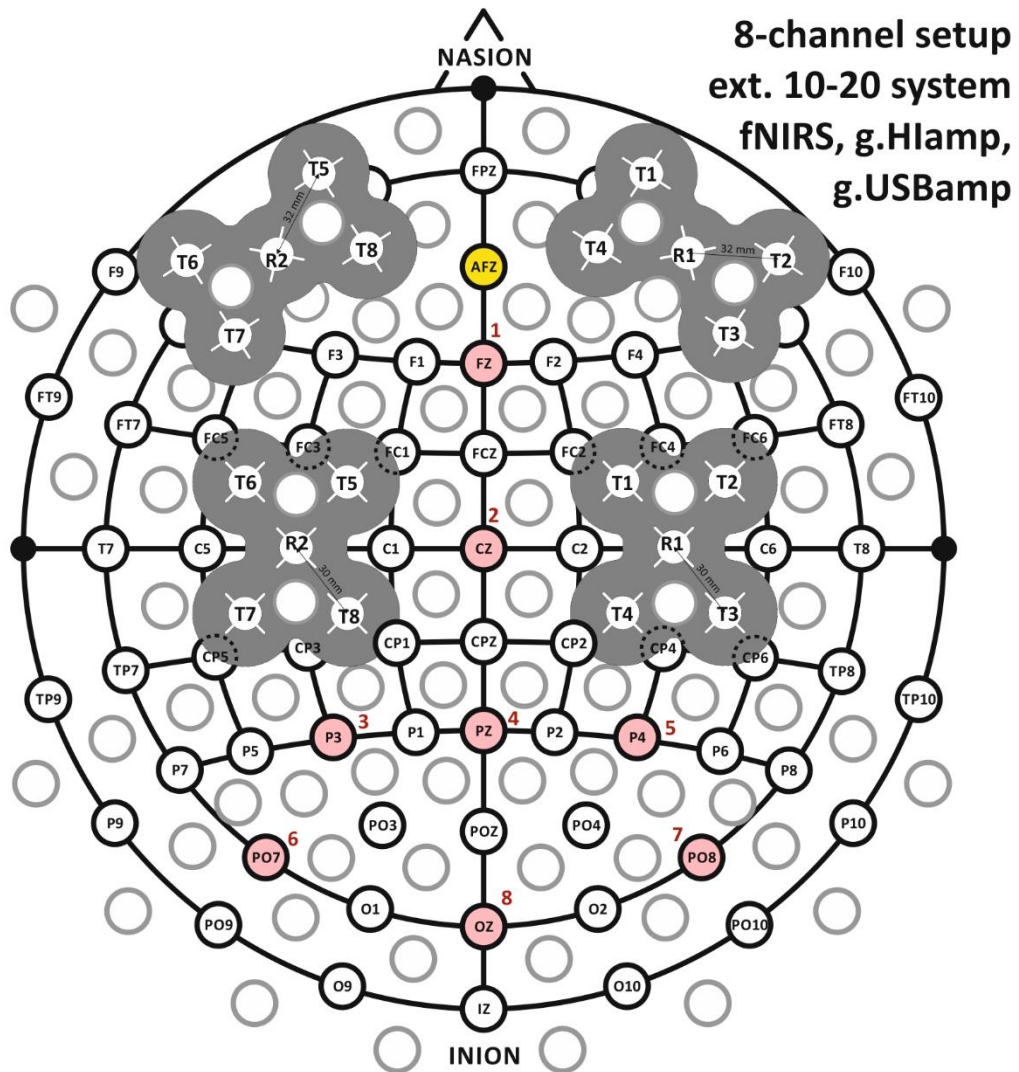




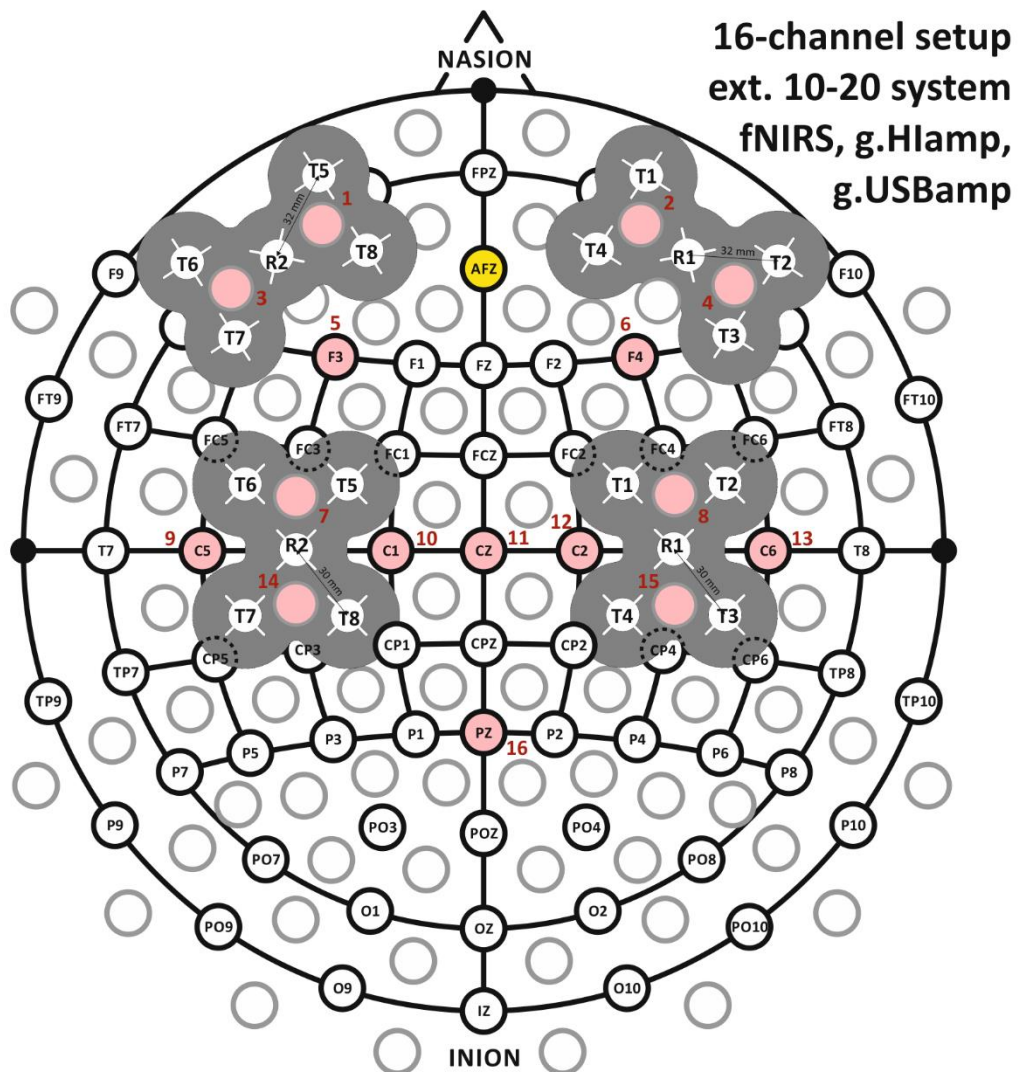


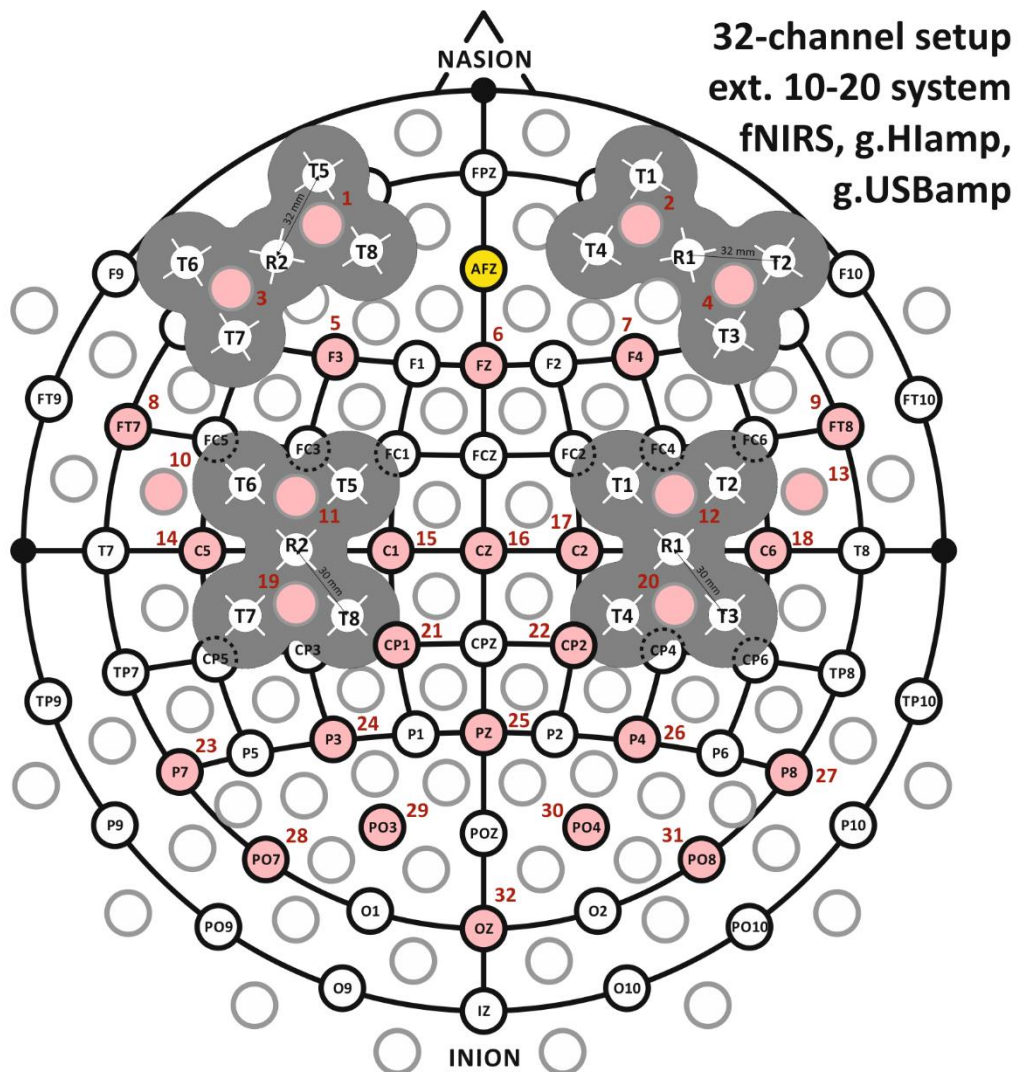


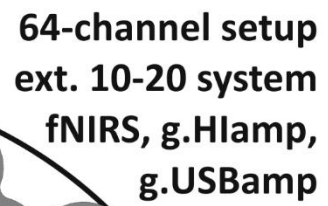
Assembling 8 channels, 8 fNIRS



Assembling 16 channels, 8 fNIRS







7 SAFE OPERATION OF G.SENSOR 8 FNIRS



CAUTION

- Use on healthy, intact skin only.
- Do not use on or near open wounds, bruised or weakened skin, whether due to injury and/or other medical conditions of the subject.
- Do not use on subjects with a history of skin allergies or sensitivity to cosmetics and lotions.
- It is not recommended that the cap be left in place for longer than eight hours for any single recording session.

Step 1: Prepare the g.SENSOR 8 FNIRS system

At delivery of the system, the accumulator may be partly discharged. Charge the accumulator. After charging, open the battery housing by pressing the cover lid and sliding it away from the front display. Insert a charged battery and close the cover.

Step 2: Optode insertion

Insert the optodes into the holder rings and choose the lowest level position. Use the low-power transmitters for the forehead spacer pads or use the high-power transmitters for the central/motor area spacer pads. Connect the optode leads to the input connectors R1, R2 (receiver optodes) and T1 to T8 (transmitter optodes). Put on the cap and assure the correct position of the cap according to the international 10-20 system.

Step 3: Optimize the optode positions for comfortable recordings

Make sure that the tips of the optodes have direct contact with the skin. For dense hair, it might be necessary to brush the hair aside with a cotton bud. In case an optode causes pain to the subject, release the optode by pressing the button at the side of the holder ring and choose a higher level position to reduce the pressure against the skin.

Step 4: Switch on the device and verify data quality

Turn on the device and connect to the recording software. Inspect the signals to assure that all optodes are placed correctly and the signal quality is good. A pulse curve signal should be visible in all the channels. Please refer to the manuals describing FNIRS data recording with High-Speed online Processing for Simulink or other recording software.



NOTE:

Please note that the high-power transmitters have a higher current consumption, which limits recording time to about 30 minutes to one hour. The battery needs to be recharged or exchanged before the next recording session.

Step 6

After finishing the recording session, turn off the device, remove the optodes from the cap and clean the entire system.

8 CLEANING THE g.SENSOR 8 FNIRS



NOTE:

The optode connector box and the optodes are not protected against liquids! Avoid any liquids or gel entering these components.

To clean the equipment, remove the optodes from the cap and disconnect them from the connector box. When washing the cap and the electrodes in water, carefully protect the optode connector box against water entering the enclosure. Please see chapter "Cleaning and disinfection of Headset, cap and electrodes".

To clean the optodes, use a damp cloth only. Never submerge the optodes in liquids! For disinfection, use disinfection wipes such as Mikrozid® sensitive wipes #70000807, Mikrozid® AF wipes #109203 from Schülke & Mayr GmbH, www.schuelke.com, or similar products.

Make sure that the entire system is completely dry before packing and storing it.

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